

Fraction word problem special project

Unit 3 · Remaining problem + Inverse towards reasoning · 65 minutes · 1-to-3 Online Lesson

Corresponding textbooks: "New Thinking in Primary School Mathematics (Second Edition)" Volume 5B Unit 3 Comprehensive + Modern Education 5L B Unit 13

Core traps: ☐ T3 misjudgment of fractional division word problems · T7 "What fraction is left" is mistaken for "fraction of the original number"

SSPA Related:  **Extremely High Frequency** Paper 1 accounts for about 20%, Paper 2 accounts for about 15%

Prerequisite knowledge: Chapter 7-8 (addition and subtraction) · Chapter 9 (multiplication) · Chapter 22 (division) · Chapter 23 (mix of four)

The goal of this class: ❶ Distinguish the operation types of fraction word problems ❷ Identify the "remaining fraction" trap ❸ Multi-step reasoning ❹ Reverse reasoning (reverse the original number from the result)

Student Name:

Class:

Date:

Time Spent:

I.Warm-Up Questions (total 4 question , 5 minutes)

#	Question	Difficulty	Working Space (Show full working)
1	Convert "A bottle of juice has $\frac{2}{3}$ liters, drink $\frac{1}{4}$ liters" into a formula. (Just list the formula, no calculation required)	Basic	
2	"Divide $\frac{3}{4}$ kilograms of candy equally among 5 people" - what operation is this? Column formula.	Basic	
3	"Used full-length SEP " - What fraction of the full length is the rest? (Tip: full length = 1) $\frac{1}{3}$ " —What fraction of the total length is the rest? (Tip: full length = 1)	Basic	
4	Are "remaining SEP " and "original SEP " necessarily the same? Give examples. $\frac{1}{2}$ " and "Original $\frac{1}{2}$ "Is it necessarily the same? Give examples.	Advanced	

II.Core Knowledge + Worked Examples

Knowledge Point 1: One-Step Fraction Problems—Identify Operation Types ● SSPA

Keywords → Operation comparison table:

- ① "total" "total" "total" → **addition**(total different parts)
- ② "remainder" "remaining" "difference" → **subtraction**(deduct a part from the total)
- ③ (Find the fraction of a quantity)"Equal share" "Each portion" "How much can be..." → Division
- ④ (distribute equally or include division)(divide equally or include division)



addition

total, total combined, total



Subtraction

Remaining, remaining difference, used



multiplication

Several times or parts of...



division

How much can be divided equally, each...

Example 1 - Addition Identification

Xiaomei has $\frac{1}{2}$ boxes of chocolates, and her mother buys $\frac{2}{3}$ boxes. How many boxes of chocolate does she have in total?

Example 2 - Subtraction Identification

A bottle of juice originally contains $\frac{5}{6}$ liters, and drank $\frac{1}{3}$ liters. How many liters are left?

#	Question	Difficulty	Working Space
5	The cake shop sold $\frac{2}{5}$ cakes in the morning and $\frac{1}{3}$ cakes in the afternoon. How many cakes were sold throughout the day?		
6	The original water in the water bottle is $\frac{3}{4}$ liters, pour out $\frac{1}{5}$ liters. How many liters are left?		
7	A rope is $\frac{2}{3}$ meters long, and the full length of $\frac{1}{2}$ is used. How many meters were used? (Note: It means "using $\frac{1}{2}$ of the full length", not "using $\frac{1}{2}$ meter"!)		
8	$\frac{3}{5}$ Kilogram of sugar, divided equally into 3 small bags. How many kilograms of sugar are in each bag?		

⚠ Q7 demonstrates the key trap: "Using $\frac{1}{2}$ of the full length" vs "Using $\frac{1}{2}$ meter" - the former is multiplication (finding the part), the latter is subtraction (deducting the length). If the judgment is wrong, the entire question will be wrong!

Knowledge point 2: "The remaining fraction" trap 🚫 SSPA extremely high frequency killer question

This is the most common question type to get wrong in the full test!

- ① "Remaining a/b" ≠ "Original a/b" - You must first find the remaining amount, and then take its fraction
- ② Solution: First find **Remaining**= Original – Used/Eat/Spent
- ③ Find again **The remaining fraction**= Remaining × Fraction
- ④ **Never multiply directly by the original number!**

"The remaining fraction" trap diagram

All

Used 1/3

Remaining 2/3

↓

Remaining

Use the remaining 1/2

Still

❌ Wrong: Directly $\frac{1}{3} + \frac{1}{2} = \frac{5}{6}$
Treat "remaining 1/2" as "all 1/2"

✅ Right: Remaining $\times \frac{1}{2} = \frac{2}{3} \times \frac{1}{2} = \frac{1}{3}$
The "remaining" must be calculated first and then multiplied

❏ **Trap detonation example 3 (the most important demonstration in this class!)**

A ribbon is $\frac{3}{4}$ meters long. After using $\frac{1}{3}$ meters, use the remaining $\frac{1}{2}$ to wrap gifts. How much rice did it take to wrap the gift?

❌ **Fatal errors (80% of students)**

$$\frac{3}{4} \times \frac{1}{2} = \frac{3}{8} \text{ rice}$$

✅ **Correct solution**

Just calculate the "remaining $\frac{1}{2}$ " as " $\frac{1}{2}$ of the original ribbon!"

$$\text{remaining} = \frac{3}{4} - \frac{1}{3} = \frac{5}{12} \rightarrow$$

$$\text{Wrapping gifts} = \frac{5}{12} \times \frac{1}{2} = \frac{5}{24} \text{ rice}$$

You must first find the remainder, and then take the remaining $\frac{1}{2}$ (not $\frac{1}{2}$ of the original ribbon!)

 **Tip: "When you see "remainder", first subtract and then multiply - first subtract the remainder and then multiply the fraction. Circle the three words "remainder" and be careful not to multiply the original number directly!"**

Example 4 - Multiplicative "What fraction is left"

Xiao Ming has pocket money |||SEP|||. After spending |||SEP|||, he saves the remaining |||SEP|||.

How much have you saved? (Answer with mixed fractions) $\frac{3}{5}$, it took $\frac{1}{5}$ Finally, put the remaining $\frac{2}{3}$ Savings. How much have you saved? (Answer with mixed fractions)

Knowledge Point 2 Synchronous practice (circle "the remaining" → first find the remaining → again take fraction → answer sentence)

#	Question	Difficulty	Working Space
9	A bottle of orange juice $\frac{7}{8}$ liters, drank $\frac{1}{4}$ liters. The remaining $\frac{1}{3}$ is used to make jelly. How many liters are used to make jelly?		
10	A bag of sugar weighs $\frac{4}{5}$ kilograms. After using $\frac{1}{5}$ kilograms, the remaining $\frac{1}{2}$ is used to make cakes. How many kilograms were used to make the cake?		
11	A rope is $\frac{5}{6}$ meters long. Cut off $\frac{1}{3}$ meters first, and use the remaining $\frac{2}{5}$ to make knots. How many meters did it take to make the knot?		
12	A box of juice $\frac{3}{4}$ liters, drank $\frac{1}{2}$ liters. Then divide the remainder equally among 3 friends. How many liters does each person receive?		
13	The sister had a box of chocolate SEP . After eating the $\frac{5}{8}$ box, the remaining $\frac{1}{4}$ was given to her brother. How many boxes does the brother get? (Note: There are two traps in the question - subtract first and then multiply!) $\frac{3}{4}$ Gave it to my brother. How many boxes does the brother get? (Note: There are two traps in the question - subtract first and then multiply!)		

 **The second biggest trap: "divide the remainder equally" = first subtract (find the remainder), then divide (divide equally). Not divide first and then subtract! In the wrong order, the answer is completely different.**

Knowledge point 3: Multi-step reasoning text questions ● SSPA advanced must test

A question contains two or more operational steps, which need to be executed in the correct order:

- ① **Circle all key data**(quantity, fraction, relational words)
- ② **Draw process arrow** |||SEP|||: **original** → **first step** (+-×÷) → **intermediate result** → **second step** (+-×÷) → **answer**|||SEP||| (Add parentheses when necessary)
- ③ **Calculate**step by step, check whether it is reasonable at each step (for example, the remaining value cannot be greater than the original)
- ④ **Calculate step by step**, check whether each step is reasonable (for example, the remainder cannot be greater than the original)

Example 5 – Addition, Subtraction and Mixing

The original water in the bucket is $\frac{5}{6}$ liters. After using $\frac{1}{3}$ liters, add $\frac{1}{2}$ liters. How many liters of water are there now? (need to subtract first and then add)

Example 6 - Mixed multiplication and division

A rope is $\frac{2}{3}$ meters long. After using the full length of |||SEP|||, divide the remaining length into 3 equal parts. How many meters long is each section? $\frac{1}{4}$ Finally, divide the remaining pieces into 3 equal parts. How many meters long is each section?

Knowledge Point 3 Synchronous practice (draw process → column → step by step calculation → answer sentence)

#	Question	Difficulty	Working Space
14	A bottle of juice $\frac{3}{4}$ liters, drank $\frac{1}{4}$ liters in the morning, and added $\frac{1}{2}$ liters in the afternoon. How many liters are there in total now?		
15	A box of cakes weighs $\frac{5}{8}$ kg. After cutting off $\frac{1}{4}$ kilograms, divide the remainder equally among 3 people. How many kilograms does each person get?		
16	Xiaohua has a $\frac{3}{5}$ premium, and Xiao Ming has $\frac{1}{3}$ 2 times the premium. How many liters of water do they have in total?		
17	A rope is $1\frac{1}{2}$ meter long. After using the full length of SEP , use the remaining $\frac{2}{5}$ for decoration. How much rice was used to make the decoration? $\frac{1}{3}$ Used for decoration. How much rice was used to make the decoration?		

Knowledge Point 4: Reverse Reasoning Word Questions ● SSPA Killer Questions

Infer the original quantity from the result - present the score question of Paper 2!

- ① **Identification** |||SEP|||: The question gives the "final result" and asks you to find "how many there

are"Strategy |||SEP|||: From the final result

② **Reverse operation**(addition, subtraction, subtraction, addition, multiplication, transformation, division, division, transformation and multiplication)**Formula:** Forward sequence formula, solve the problem in reverse direction - reverse each step in the forward direction

③ **Check calculation:** Use the calculated original number, calculate it once in the forward direction, it should be equal to the result given in the question

④ **Check calculation:** Use the calculated original number and calculate it once, it should be equal to the result given in the question.



Thinking in the direction

original $\rightarrow$ subtraction $\rightarrow$
remainder $\rightarrow$ $\times$ fraction



reverse operation

Infer from the result that $\div$
becomes $\times$, $+$ becomes $-$



Check calculation

Use the original number to
calculate and confirm



trap

Pay attention to unit and
fraction types

Example 7 - Reverse Reasoning (Subtraction Reverse)

Xiao Ming had some chocolates. After eating $\frac{1}{4}$ boxes, there were $\frac{1}{2}$ boxes left. How many boxes of chocolate were there?

Example 8 - Reverse Reasoning (Hybrid Reverse)

From a rope, first cut off $\frac{1}{3}$ meters, and then use the remaining $\frac{1}{2}$ for handicrafts. For handicrafts, $\frac{1}{4}$ meters are used. How many meters long was the rope?

Knowledge Point Four Synchronization Practice

#	Question	Difficulty	Working Space
18	A bottle of juice, after drinking $\frac{1}{3}$ liters, there will be $\frac{1}{4}$ liters left. How many liters did it originally have?		
19	Some candies are divided equally among 4 people, and each person gets $\frac{1}{6}$ kilograms. How many kilograms of candy were there?		
20	A box of cakes, eaten $\frac{1}{2}$ boxes, and then eaten $\frac{1}{3}$ boxes, leaving $\frac{1}{6}$ boxes. How many boxes of cakes were there?		

Tip (reverse reasoning): "Calculate backwards from end to beginning, add, subtract, multiply, and divide, subtract, add, divide, multiply, and remember to check it after you finish the calculation."

III. Lesson Layered Synchronization Practice

Basic layer (total 5 questions, all must do, all text questions)

#	Question	Difficulty	Working Space
21	Xiaomei has $\frac{2}{3}$ meters of ribbon, and Xiaohua has $\frac{1}{4}$ meters. How many meters of ribbon do they have in common?		
22	A bottle of water has $\frac{3}{5}$ liters and uses $\frac{1}{10}$ liters. How many liters are left?		
23	The $\frac{1}{3}$ of the cake is chocolate flavor, and the rest is strawberry flavor. What percentage of the cake is strawberry flavor?		
24	$\frac{3}{4}$ Kilograms of flour, make one cake per $\frac{1}{8}$ kg. How many cakes can be made?		
25	A rope is $\frac{4}{5}$ meters long. After using $\frac{1}{5}$ meters, the remaining $\frac{1}{2}$ How many meters have been left?		

Advanced layer (total 4 questions,   choose do)

#	Question	Difficulty	Working Space
26	A bottle of juice $\frac{5}{6}$ liters. After pouring $\frac{1}{2}$ liters, divide the remaining juice equally among 4 people. How many liters does each person get?		
27	A bag of rice weighs $\frac{7}{8}$ kg. $\frac{1}{4}$ kilograms were used to cook rice, and the remaining $\frac{2}{3}$ were used to cook porridge. How many kilograms were used to cook the porridge?		
28	My brother used $\frac{1}{3}$ of his pocket money to buy books, and then used the remaining $\frac{1}{2}$ to buy stationery. He spent $\frac{10}{1}$ yuan to buy stationery. How many yuan was it originally?		
29	There are $\frac{3}{5}$ liters of juice in the bottle. After pouring $\frac{1}{4}$ liters of water, pour out $\frac{1}{3}$ of the mixture. How many liters were poured out?		

 challenge layer (total 2 questions,   choose do, SSPAKiller Questions)

#	Question	Difficulty	Working Space
30	For a rope, first cut off SEP , then cut off the remaining SEP , and finally there will be 3 meters left. How many meters long was the rope? (First find what fraction of		

	the original amount is left after the second cut, and then work in reverse) $\frac{1}{3}$, and then cut off the rest $\frac{1}{2}$, and finally 3 meters remain. How many meters long was the rope? (First find what fraction of the original amount is left after the second cut, and then work in reverse)		
31	Xiao Ming spent $\frac{2}{5}$ of his pocket money to buy snacks, and then spent the remaining $\frac{1}{3}$ to buy stationery, and finally he was left with $\frac{4}{1}$ yuan. How much is the original pocket money?		

IV. The ultimate challenge — SSPA Paper 2 practical question

#	Question	Difficulty	Working Space
 1	SEP on the bookshelf is for Chinese books, $\frac{1}{4}$ is for English books, and the remaining $\frac{1}{3}$ is for math books. If mathematics books occupy SEP , what fraction of the total space on the bookshelf is occupied by books? (Check: Chinese + English + Mathematics = total number of books) $\frac{1}{2}$ Put the math book. If math books took up bookshelf $\frac{5}{24}$, what fraction of the total space on the bookshelf is occupied by books? (Check: Chinese + English + Mathematics = total number of books)		
 2	A barrel of oil contains $\frac{5}{6}$ liters. The first time I used $\frac{1}{4}$ liters, the second time I used the remaining SEP , and the third time I used the remaining $\frac{2}{3}$ of the second time. How many liters are left at the end? $\frac{1}{2}$. How many liters are left at the end?		

Comprehensive reasoning question (Choose do, Paper 2 points question)

#	Question	Difficulty	Working Space
R1	For a bottle of water, I used $\frac{1}{4}$ on the first day, used the remaining $\frac{1}{3}$ on the second day, and ended up with $\frac{1}{2}$ liters left. How many liters of water was there? (Hint: The remaining $\frac{1}{3}$ of the first day was used on the second day, that is, $\frac{1}{3}$ of the "original $\frac{3}{4}$ " was used on the second day = the original $\frac{1}{4}$)		
R2	A, B, and C share candies. A ate SEP , B ate the remaining SEP , and C ate the last remaining box (SEP box). How many boxes of candy were there? $\frac{1}{3}$, B ate the rest $\frac{1}{2}$, B ate all that was left ($\frac{1}{4}$ box). How many boxes of candy were there?		

V. Class afterhomework

Basic must-do questions (total 6 questions, all text questions, must write answer sentences)

#	Question	Difficulty	Working Space
H1	A bottle of orange juice $\frac{3}{4}$ liters, my brother drank $\frac{1}{4}$ liters, and my sister drank $\frac{1}{3}$ liters. How many liters did they drink in total?		
H2	A rope is $\frac{4}{5}$ meters long, cut off $\frac{1}{10}$ meters. How many meters are left?		
H3	$\frac{2}{3}$ Kg of sweets, one pack per $\frac{1}{6}$ kilogram. How many bags can be packed?		
H4	A pancake, the sister ate $\frac{1}{3}$ and the brother ate SEP . How much of the pancake is left? $\frac{1}{4}$. How much of the pancake is left?		
H5	A bag of sugar weighs $\frac{5}{8}$ kilograms. After using $\frac{1}{4}$ kilograms, the remaining $\frac{1}{2}$ is used to make desserts. How many kilograms were used to make the dessert?		
H6	A ribbon is $\frac{3}{5}$ meters long, and the full length of $\frac{1}{3}$ is used. How many meters were used?		

For advanced, choose doquestion (total 2 questions,  choose do)

#	Question	Difficulty	Working Space
H7	A rope is $1\frac{1}{2}$ meter long. After using $\frac{1}{2}$ meters, the remaining $\frac{1}{4}$ is used to make knots. How many meters did it take to make the knot?		
H8	Xiao Ming had pocket money, so he spent $\frac{1}{3}$ to buy stationery, and then spent the remaining $\frac{1}{2}$ to buy snacks, and finally he was left with SEP . How much is the original pocket money? (Expressed as a fraction, assuming original pocket money = 1) $\frac{1}{3}$. How much is the original pocket money? (Expressed as a fraction, assuming original pocket money = 1)		

VI. The Lessoncorecommon errorssummary

☒ Self-examination in this hall (tick after completion)

☐ I know the pitfalls of solving each knowledge point

☐ I can complete  basic questions independently

☐ I can challenge  advanced questions

☐ I remember the formula

 Review of Learning Objectives - After completing this lesson you should be able to:

☐ Identify all trap types in our hall

☐ Solve  basic questions independently (100% correct)

☐ Challenge  Advanced questions (80%+ correct)

☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	"Used $\frac{1}{2}$" vs "Used $\frac{1}{2}$ meter" confusion SEP : The former is a multiple (multiplication), the latter is a quantity (subtraction): The former is a multiple (multiplication), the latter is a quantity (subtraction)	See clearly whether it is "a fraction of the total length" or "how many meters". "" is mostly multiplication.
2	"Remaining a/b" is calculated as "original a/b"	You must first find the remaining quantity, and then take the remaining a/b! This is the most important lesson in the whole class.
3	Direct addition and subtraction of fractions of the remaining amount SEP : The remaining $\frac{1}{2}$ and the remaining $\frac{1}{3}$ are directly added: The remaining $\frac{1}{2}$ and the remaining $\frac{1}{3}$ are added directly	The "remainder" is based on a different basis each time, and must be calculated individually and then added together.
4	The order of multi-step questions is wrong (Add first and then multiply as multiplication first and then add)	Circle keywords and draw process arrows to confirm the calculation order.
5	Inverse questions will not be "operated backwards"	Forward continuation, solve the problem in reverse: + changes to -, - changes to +, $\times$ changes to $\div$, $\div$ changes to $\times$
6	Missing sentences/missing units	Text questions must write "Answer:..." + complete unit, otherwise step points will be deducted
7	The fraction result is unreduced	The final step must check reduction (HCF) and improper fractions $\rightarrow$ mixed numbers